

ANSWER KEY — Integration Lesson 20 Test: Capstone, Putting It All Together

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. (adding rate $\times$ time pieces) You bike at 10 mph for 2 hours, then at 14 mph for 1.5 hours. What is your total distance?

Work: chop—multiply—add (Lesson 1): $10 \times 2 = 20$ miles; $14 \times 1.5 = 21$ miles; total = $20 + 21$.

Answer: 41 miles.

2. (area under a graph) A train's speed rises in a straight line from 4 m/s to 10 m/s over 6 seconds. Find the distance by geometry.

Work: distance = area under the speed graph (Lesson 2). Trapezoid: $\frac{1}{2} \times (4 + 10) \times 6 = \frac{1}{2} \times 14 \times 6 = 42$. (Or: rectangle $4 \times 6 = 24$ plus triangle $\frac{1}{2} \times 6 \times 6 = 18$; $24 + 18 = 42$ ✓.) **Answer: 42 meters.**

3. (Fundamental Theorem) Evaluate $\int$ from 1 to 3 of $(2x + 1) dx$.

Work: antiderivative $F(x) = x^2 + x$. FTC: $F(3) - F(1) = (9 + 3) - (1 + 1) = 12 - 2$. **Answer: 10.**

4. (signed area: net vs total) $v(t) = 4 - 2t$ m/s for $0 \leq t \leq 4$. Find (a) net displacement and (b) total distance.

Work: (a) net = $\int_0^4 (4 - 2t) dt = [4t - t^2]_0^4 = 16 - 16 = 0$ m. (b) $v = 0$ at $t = 2$. Forward: $\int_0^2 (4 - 2t) dt = [4t - t^2]_0^2 = 8 - 4 = 4$ m. Backward: $\int_2^4 (4 - 2t) dt = [4t - t^2]_2^4 = 16 - 16 - (8 - 4) = -4$ m, i.e. 4 m backward. Total = $4 + 4 = 8$.

Answer: (a) net = 0 m (it returns to its start) · (b) total = 8 m.

5. (area between curves) Find the area between $y = x$ and $y = x^2$ (crossings at $x = 0$ and 1).

Work: on $[0, 1]$ the line is on top (at $x = \frac{1}{2}$: $\frac{1}{2}$ vs $\frac{1}{4}$). Area = $\int_0^1 (x - x^2) dx = [x^2/2 - x^3/3]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{3}{6} - \frac{2}{6}$. **Answer: 1/6 square unit.**

6. (gravity, $g = 10$) A ball is thrown straight up from the ground at 30 m/s. When is the peak, and how high?

Work: $v(t) = 30 - 10t = 0 \rightarrow t = 3$ s. Height = $\int_0^3 (30 - 10t) dt = [30t - 5t^2]_0^3 = 90 - 45 = 45$.

Answer: peak at $t = 3$ seconds, 45 meters high.

7. (average value) Find the average value of $f(x) = x^2$ on $[0, 3]$.

Work: average = $(1/(b - a)) \int_a^b f = (1/3) \int_0^3 x^2 dx = (1/3) \times [x^3/3]_0^3 = (1/3) \times 9 = 3$. **Answer: 3.** (Sensible: f runs from 0 to 9; the average leans low because x^2 spends more time small.)

8. (e^x and $1/x$) (a) $\int$ from 0 to 1 of $e^x dx$ and (b) $\int$ from 1 to e of $(2/x) dx$.

Work: (a) $[e^x]_0^1 = e - 1 \approx 2.718 - 1 = 1.718 \approx 1.72$. (b) $[2 \ln x]_1^e = 2 \ln e - 2 \ln 1 = 2(1) - 2(0) = 2$. **Answer: (a) $e - 1 \approx 1.72$ • (b) 2.**

9. (linear inside) Evaluate $\int$ from 0 to $\pi/4$ of $\cos(2x) dx$.

Work: linear inside ($a = 2$): antiderivative = $\sin(2x)/2$. Evaluate: $\sin(\pi/2)/2 - \sin(0)/2 = 1/2 - 0$.
Answer: $1/2$. (Common error: forgetting the $\div 2$ gives 1, wrong.)

10. (u-substitution) Evaluate $\int$ from 0 to 1 of $2x(x^2 + 1)^3 dx$.

Work: $u = x^2 + 1$, $du = 2x dx$ — the $2x dx$ is exactly present. Limits: $x = 0 \rightarrow u = 1$; $x = 1 \rightarrow u = 2$.
Integral = $\int_1^2 u^3 du = [u^4/4]_1^2 = (16 - 1)/4 = 15/4$. **Answer: $15/4 = 3.75$.**